

Lecture 11: Computer Security

◆ Computer Security

- Protection against damage or theft of hardware, software, and information.
 - Prevents threats from viruses, malware, and unauthorized use.
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◆ Cybercrime

- Crimes committed using computers and networks.
 - Examples: hacking, phishing, online fraud.
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◆ Hackers vs Crackers

- **Hacker:** Skilled person who identifies and exploits network loopholes.
 - **Cracker:** Gains unauthorized access to systems with malicious intent.
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◆ Phishing

- Fake emails or websites used to steal personal information.
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◆ Cyberbullying & Harassment

- Using electronic means (computers, mobiles, internet) to threaten or harass.
 - Victims should report to trusted people or authorities.
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◆ Cyber-Attack

- Criminals use devices to attack single or multiple networks.
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◆ Malware

- Malicious software: viruses, worms, spyware, adware.
 - Spread via USBs, CDs, downloads, networks, email attachments.
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◆ Antivirus Software

- Utility software that protects computers from malware and viruses.
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◆ Data Safety

- Important files should be backed up in multiple locations.
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◆ Authentication Mechanisms

- Hardware/software systems that ensure only authorized users access data.
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◆ Professional Ethics

- Principles and rules guiding professional behavior in computing.
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◆ Information Accuracy

- Ensures information is correct and reliable.
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◆ Intellectual Property Rights (IPR)

- Protects intangible creations (software, designs, writings).
 - Enforced through copyrights, patents, trademarks.
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◆ Software Piracy

- Illegal use, copying, or distribution of copyrighted software.
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◆ Plagiarism

- Presenting someone else's work or ideas without proper acknowledgment.
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■ MCQs – Computer Security

1.

Computer security mainly protects:

- A) Only hardware
 - B) Only software
 - C) Hardware, software, and information
 - D) Only networks
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2.

Cybercrime refers to:

- A) Physical theft of computers
 - B) Crime committed using computers and networks
 - C) Software piracy only
 - D) Plagiarism
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3.

A hacker is someone who:

- A) Creates viruses
 - B) Uses skills to find and exploit loopholes
 - C) Copies software illegally
 - D) Sends phishing emails
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4.

Crackers are:

- A) Ethical hackers
 - B) People who gain unauthorized access to systems
 - C) Antivirus developers
 - D) Network administrators
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5.

Phishing is done through:

- A) Fake emails and websites
 - B) USB drives
 - C) Antivirus software
 - D) Firewalls
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6.

Cyberbullying involves:

- A) Physical bullying in schools
 - B) Threats or harassment using electronic means
 - C) Copying software illegally
 - D) Stealing hardware
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7.

Victims of cyber harassment should:

- A) Ignore it
 - B) Report to trusted people and authorities
 - C) Delete their accounts silently
 - D) Do nothing
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8.

A cyber-attack is when:

- A) A virus spreads through USB
 - B) A criminal uses devices to attack networks
 - C) A student copies assignments
 - D) A company backs up data
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9.

Malware is:

- A) Utility software
 - B) Malicious software like viruses, worms, spyware
 - C) Antivirus software
 - D) Backup software
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10.

Viruses can spread through:

- A) USB drives and CDs
 - B) Internet downloads
 - C) Email attachments
 - D) All of the above
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11.

Antivirus software is designed to:

- A) Create viruses
 - B) Protect computers from malware
 - C) Copy files
 - D) Increase storage capacity
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12.

For data safety, users should:

- A) Avoid using computers
 - B) Back up important files in multiple places
 - C) Delete files regularly
 - D) Use only one storage device
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13.

Authentication mechanisms ensure:

- A) Only authorized users access data
 - B) Viruses are removed
 - C) Files are backed up
 - D) Software piracy is stopped
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14.

Professional ethics guide:

- A) Hardware design
 - B) Behavior in a profession
 - C) Antivirus installation
 - D) Network speed
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15.

Information accuracy ensures:

- A) Data is correct and true
 - B) Data is encrypted
 - C) Data is compressed
 - D) Data is backed up
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16.

Intellectual property refers to:

- A) Tangible assets
 - B) Intangible creations of human intellect
 - C) Hardware devices
 - D) Computer viruses
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17.

Intellectual Property Rights protect:

- A) Hardware only
 - B) Software only
 - C) Copyrights, patents, trademarks
 - D) Only antivirus software
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18.

Software piracy means:

- A) Legal use of software
 - B) Illegal copying or distribution of software
 - C) Installing antivirus
 - D) Encrypting data
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19.

Plagiarism means:

- A) Copying someone's work without acknowledgment
 - B) Creating original work
 - C) Encrypting files
 - D) Backing up data
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20.

Which of the following is NOT malware?

- A) Worms
 - B) Spyware
 - C) Adware
 - D) Antivirus
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21.

Phishing emails usually try to steal:

- A) Hardware devices
 - B) Personal information like passwords
 - C) Antivirus software
 - D) Storage capacity
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22.

Cybercrime can include:

- A) Hacking
 - B) Phishing
 - C) Online fraud
 - D) All of the above
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23.

Which of the following is a preventive measure against malware?

- A) Installing antivirus software
 - B) Ignoring updates
 - C) Sharing passwords
 - D) Using pirated software
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24.

Unauthorized access to systems is done by:

- A) Crackers
 - B) Antivirus software
 - C) Backup systems
 - D) Ethical hackers
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25.

Which of the following is an ethical responsibility in computing?

- A) Plagiarism
 - B) Professional ethics
 - C) Software piracy
 - D) Cyber-attack
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26.

Viruses can be prevented by:

- A) Using strong passwords
 - B) Installing antivirus and avoiding suspicious downloads
 - C) Ignoring updates
 - D) Sharing USB drives freely
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27.

Cyber-attack targets:

- A) Single or multiple networks
 - B) Only hardware
 - C) Only software
 - D) Only storage devices
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28.

Which of the following is a malicious activity?

- A) Backup of files
 - B) Phishing
 - C) Authentication
 - D) Professional ethics
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29.

Copyrights, patents, and trademarks are part of:

- A) Intellectual Property Rights
 - B) Antivirus software
 - C) Authentication mechanisms
 - D) Backup systems
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30.

The main goal of computer security is:

- A) To protect against damage, theft, and unauthorized use
 - B) To increase internet speed
 - C) To reduce storage capacity
 - D) To copy software freely
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✓ Answers with Explanations

1. **C – Hardware, software, and information** → Computer security protects all.

2. **B – Crime using computers/networks** → Cybercrime definition.
3. **B – Finds and exploits loopholes** → Hacker role.
4. **B – Unauthorized access** → Crackers definition.
5. **A – Fake emails/websites** → Phishing method.
6. **B – Threats via electronic means** → Cyberbullying.
7. **B – Report to authorities** → Correct response.
8. **B – Criminal attacks networks** → Cyber-attack definition.
9. **B – Malicious software** → Malware includes viruses, worms, spyware.
10. **D – All of the above** → Viruses spread through multiple sources.
11. **B – Protect computers** → Antivirus role.
12. **B – Backup files** → Data safety measure.
13. **A – Authorized access only** → Authentication purpose.
14. **B – Behavior in profession** → Ethics definition.
15. **A – Correct and true data** → Information accuracy.
16. **B – Intangible creations** → Intellectual property.
17. **C – Copyrights, patents, trademarks** → IPR protection.
18. **B – Illegal copying/distribution** → Software piracy.
19. **A – Copying without acknowledgment** → Plagiarism.
20. **D – Antivirus** → Not malware.
21. **B – Personal info** → Phishing goal.
22. **D – All of the above** → Cybercrime includes all.
23. **A – Installing antivirus** → Preventive measure.
24. **A – Crackers** → Unauthorized access.
25. **B – Professional ethics** → Ethical responsibility.
26. **B – Antivirus + safe downloads** → Virus prevention.
27. **A – Networks** → Cyber-attack targets.
28. **B – Phishing** → Malicious activity.
29. **A – Intellectual Property Rights** → Protects creations.
30. **A – Protection against threats** → Goal of computer security.